MIL-C-22931/16
10 February 1968
SUPERSEDING
MIL-C-22931/4A (in part)
27 March 1964

MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, SEMIRIGID, COAXIAL, SEMI-AIR-DIELECTRIC 3.125 INCHES OUTSIDE DIAMETER, 75 OHMS

The complete requirements for procuring cables described herein shall consist of this document and the latest issue of Specification MIL-C-22931.

REQUIREMENTS:

Operating temperature range: -55 to +80° C.

Inner conductor: Copper wire or tube.
Overall diameter: 0.732 (18.59 mm) ± 0.002 (.05 mm) inch.

Dielectrics:
Cable core and insulating layer: Polystyrene.

Outer conductor: Aluminum tubing, smooth, seamless;
Inside diameter: 2.850 (72.39 mm) inches.
Outside diameter: 3.125 (79.38 mm) inches.

Jacket: See table.

Voltage withstand:
10,000 volts rms (nominal).
18,000 volts dc.

Insulation defects, spark: See table.

Attenuation (decibels (dB) per 100 ft maximum):
0.068 dB at 30 MHz.
0.33 dB at 400 MHz.
0.56 dB at 1,000 MHz.

Velocity: 92 percent, nominal.

Capacitance: 14.7 pf/ft, nominal.

Impedance: 75 ± 1.5 ohms.

Voltage standing wave ratio:
Initial - 1.12 maximum for 10-1,000 MHz.
Following bending and temperature cycling tests - 1.15 maximum for 10-1,000 Mhz.

Mandrel size for cold bend, bending, and temperature cycling tests - 70 (1778 mm) inches.

Minimum bending radius: 50 (1270 mm) inches.

FSC 6145

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Requirement 1/	M22931/16 dash number	
	-001	-002
Jacket - - - - - - - - - - -	Not applicable	Polyethylene
Diameter - - - - - - - - -		3.275(83.19) +0.030(.76) -0.018(.46)
Minimum wall thickness - - - -		.070(1.78)
Insulation defects, spark - - - -	Applicable	Applicable

1/ Dimensions are in inches, millimeters are in parentheses. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

NOTES:

1. Cross reference information is as follows:

Part number	RG number	Former classification Class	Type
M22931/16-001	RG-250/U	N	I
M22931/16-002	RG-251/U		

2. Figure is for information.

Custodians:
Army - EL
Navy - EC
Air Force - 11

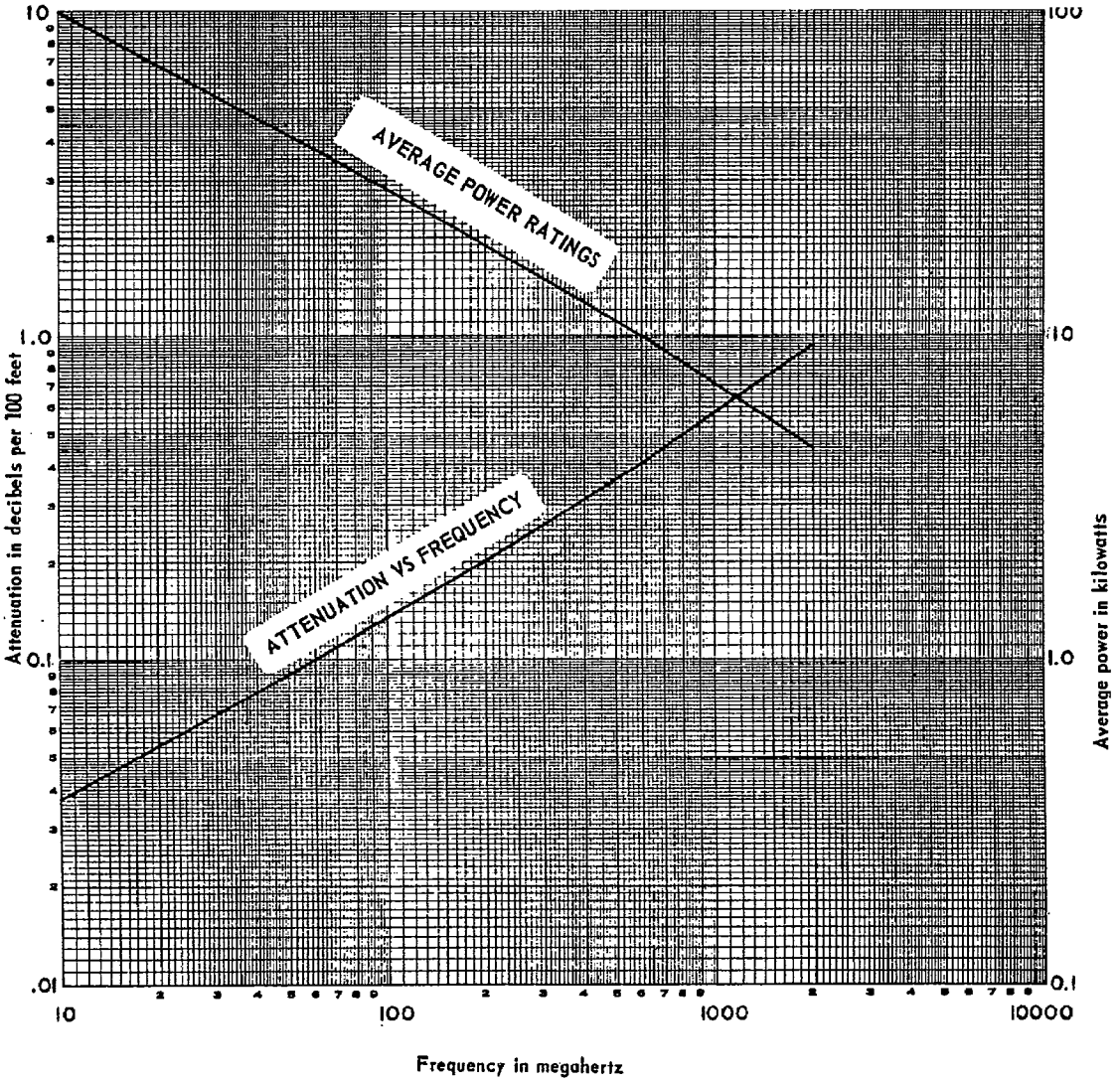
Review activities:
Army - EL, MU
Navy - SH, EC
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME, AT, AV
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Navy - EC

(Project 6145-0306-2)

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Power rating based upon a 40°C. ambient with a 45°C rise in center conductor temperature.

FIGURE 1. Attenuation and power rating curves for –001 and –002.

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